

BIPIN KUMAR OJHA

Senior Backend Engineer & Tech Lead | Cloud-Native Microservices • Golang • Distributed Systems • FinTech

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PROFESSIONAL SUMMARY

Senior Backend Engineer & Tech Lead with **19+ years** of engineering experience architecting cloud-native microservices, high-throughput distributed systems, and real-time streaming platforms. Deep expertise transitioning from enterprise Java/Spring Boot to high-performance **Golang** and **Python (FastAPI)** microservices with production deployments on **GCP and AWS** using Docker and Kubernetes. Proven record engineering **fintech/banking platforms** with ACID guarantees, Zero-Trust security, AI-powered fraud detection, and immutable audit ledgers, as well as identity resolution platforms, telemetry pipelines, and telecom solutions leveraging Google Cloud Spanner, GCP Pub/Sub, Apache Beam, MongoDB, and Redis under stringent FCRA data governance. Strong culture of quality: Go race detector, SAST (GoSec), SonarQube, vulnerability scanning (govulncheck), and end-to-end automated test suites.

CORE COMPETENCIES & TECHNICAL SKILLS

Languages	Golang, Python (FastAPI, AsyncIO, Pydantic), Java/Spring Boot, C/C++, JavaScript; gRPC, RESTful APIs, Protobuf v3, SS7
Distributed Systems	Microservices Architecture, Event-Driven Architecture, Transactional Outbox Pattern, ACID Multi-Document Transactions, Concurrency Patterns (Worker Pools, Goroutines, Channels), Idempotency, Async IPC
AI & ML	Gemini AI (generative-ai-go SDK), Agentic AI Workflows, Real-Time Fraud Detection, Predictive Analytics, LLM Integration, Fail-Open AI Architectures
Security	Zero-Trust mTLS, HMAC-SHA256 JWT, OAuth2/OIDC (Keycloak), RBAC, IDOR Protection, Rate Limiting, FCRA Compliance, GoSec SAST, govulncheck
Observability	OpenTelemetry (OTel W3C Distributed Tracing), Prometheus, Grafana, Structured Logging, Correlation IDs, RabbitMQ Dead Letter Queues
Streaming & Messaging	GCP Pub/Sub, Apache Beam, Kafka, RabbitMQ (AMQP, Quorum Queues), MQTT, AWS IoT Core
Databases	Google Cloud Spanner, MongoDB 7.0 (Replica Set, Aggregation Pipelines), PostgreSQL, Redis (Caching & State), MySQL, Google Cloud Storage (GCS)
Cloud & DevOps	GCP, AWS, Docker, Kubernetes (HPA, PDB, Ingress, ConfigMaps), GitHub Actions CI/CD, GitLab CI, Jenkins, SonarQube, GitHub Copilot
Domain Expertise	FinTech / Core Banking, Double-Entry Bookkeeping, Multi-Region Disaster Recovery, Identity Resolution, Telecom Signaling (SS7, STP, AAA, SMS Gateway)

PROFESSIONAL EXPERIENCE

Senior Go / Backend Developer — Altzor Digital Solutions India Pvt. Ltd. Mar 2026 – Present

Client: Equifax Inc. | Platform: Keying & Linking (K&L) Identity Resolution Platform | GCP

- High-Concurrency Identity Resolution (EKS Module):** Engineered distributed Go microservices with worker pools, goroutines, and channels executing compute-intensive identity-matching algorithms against massive consumer datasets in **Google Cloud Spanner**; sustained **>10,000 identity lookups/sec** under peak load with P99 latency under 50 ms.
- Real-Time Event Streaming & Async IPC:** Built asynchronous event-driven pipelines using **GCP Pub/Sub and Apache Beam** to decouple microservice interactions and process continuous address-linking streams at ultra-low latency, reducing cross-service coupling by eliminating synchronous RPCs from critical paths.
- Address Library & GCS Integration:** Developed low-latency **gRPC and REST APIs** in Go for the Address Library module, automating real-time address normalization and standardizing bulk file ingestion workflows with Google Cloud Storage (GCS), cutting manual processing time by ~60%.
- EDS Purposing View & FCRA Governance:** Designed backend policy-enforcement layers in Go to query Cloud Spanner and enforce granular data-purposing access controls, ensuring full compliance with **FCRA** and statutory data-handling regulations across all consumer data access patterns.

- **Database Optimization & CI/CD:** Optimized Spanner SQL queries, secondary indexes, and transactional read/writes to maximize throughput across globally distributed tables; automated unit/integration test suites in **Jenkins CI/CD**, achieving **>80% unit test coverage**.
- **AI-Assisted Development:** Leveraged **GitHub Copilot** to accelerate engineering velocity, generating comprehensive Go table-driven test suites, edge-case mocks, and Spanner SQL patterns; reduced feature delivery cycles by ~30%.

Senior Backend Engineer (Python / FastAPI & Golang) — QuantiqueMinds Pvt. Ltd.

Sep 2025 – Mar 2026

Domains: Workspace Management SaaS & Restaurant Management SaaS | AWS

- **High-Throughput Async APIs:** Architected and delivered high-throughput backend microservices using **Python (FastAPI)** and Golang for multi-tenant SaaS platforms; built non-blocking RESTful APIs with Pydantic and AsyncIO integrated with **MongoDB aggregation pipelines** and **Redis** delivering sub-millisecond multi-tenant caching, sustaining **5,000+ concurrent API requests**.
- **IoT Telemetry Ingestion Engine:** Engineered an IoT device management and telemetry ingestion pipeline using **Python + MQTT / Message Queues and AWS IoT Core**, processing streaming sensor data (temperature, occupancy, energy) for a fleet of 500+ IoT devices.
- **Agentic AI Integration:** Integrated **Agentic AI / predictive data services** in Python to analyze real-time telemetry—space utilization, temperature trends, and energy consumption forecasting—enabling proactive operational decisions and reducing energy waste by an estimated 20%.
- **Containerized Deployments:** Containerized all microservices with **Docker and Kubernetes**, streamlining automated deployments on AWS through full CI/CD pipelines, reducing deployment lead time from hours to minutes.

Senior Software Engineer — Kenshi Labs Pvt. Ltd.

Jul 2024 – Aug 2025

Product: SmartSME Accounting & Inventory Cloud Platform | Golang Microservices

- Developed the **SmartSME** cloud accounting and inventory platform backend from scratch in **Golang** as resilient, independently deployable microservices with clear bounded contexts.
- Optimized transactional data workflows using **MongoDB aggregation pipelines** and **Redis** caching for high query concurrency, reducing average query latency by 40%.
- Containerized backend services with **Docker**, standardizing development environments and deployment pipelines across the engineering team.

Senior Software Engineer / Team Lead — Monty Mobile

Apr 2015 – Apr 2024

- Led backend engineering for the **Go-STP distributed telecom signaling platform**, improving packet throughput by 35% and fault tolerance through resilient distributed state machines in **Golang and C++**.
- Architected **Multi-IMS I products** in Golang/C++ for global roaming, managing complex distributed telecom protocol state machines (SS7, STP) across multi-carrier networks.
- Engineered secure backend services using **PostgreSQL and Keycloak** for centralized authentication and fine-grained access control (OAuth2/OIDC).
- Mentored a team of 8+ junior and mid-level engineers; established code review benchmarks and drove key architectural decisions across multiple concurrent projects.

SS7 Software Engineer / Module Lead — XIUS (Megasoft Ltd.)

Mar 2006 – Apr 2015

- Stabilized and scaled carrier-grade **SMS Gateway backends**, sustaining high-volume message delivery under extreme peak loads for Tier-1 telecom operators.
- Led a 5-member engineering team to design distributed backend components for **roaming and AAA telecom servers**.

Software Engineer — Sai Info System (India) Pvt. Ltd.

Apr 2003 – Mar 2006

- Developed enterprise web modules and services using **Java and PHP** across diverse client engagements.

KEY PROJECTS

Global Multi-Currency Digital Wallet & Ledger Service (Personal / Open Source) NEW

Golang, MongoDB 7.0 (ACID Replica Set), gRPC + Protobuf v3, RabbitMQ, JWT/mTLS, Gemini AI, OpenTelemetry, Prometheus, Grafana, Docker, Kubernetes, GitHub Actions CI/CD

github.com/bkojha74/global-wallet-microservices

- **Production-grade fintech platform** with 5 independently deployable Go microservices: API Gateway, Wallet Service (Primary + Standby Hot DR), Ledger Service, Auth Service, and Logging Service.
- **Strict Financial Consistency:** Multi-document ACID transactions via MongoDB Replica Set (writeConcern.Majority, readConcern.Snapshot) preventing overdrafts, race conditions, and phantom balance anomalies.
- **Transactional Outbox Pattern:** Eliminates synchronous cross-service RPCs inside database transactions; guaranteed at-least-once ledger delivery via asynchronous outbox relay workers.
- **True Double-Entry Bookkeeping & SHA-256 Cryptographic Audit Chain:** Every debit/credit is journaled as an immutable, tamper-proof ledger entry with hash chaining for forensic auditability.
- **Gemini AI Real-Time Fraud Detection:** Integrated Google Gemini (gemini-3.1-flash-lite) to evaluate contextual fraud risk on every transfer before database execution; transactions with risk score >0.60 are blocked (HTTP 403). Fail-Open architecture guarantees availability during AI provider outages.
- **Zero-Trust 6-Layer Security Pipeline:** HMAC-SHA256 JWT validation, IDOR ownership enforcement, RBAC admin gating, mTLS for all internal gRPC, token-bucket rate limiting (60 rps/100 burst), Security Headers (HSTS, CSP, X-Frame-Options: DENY).
- **Multi-Region Active-Standby Disaster Recovery:** API Gateway FailoverCoordinator atomically switches routing between us-east-1 Primary and eu-west-1 Standby with zero service interruption.
- **OpenTelemetry W3C Distributed Tracing & Observability:** Correlation IDs propagated across all gRPC metadata and HTTP headers; Prometheus metrics scraped from all services; preconfigured Grafana dashboards; centralized structured logging via RabbitMQ with local disk-spooling fallback.
- **9-Stage GitHub Actions CI/CD Pipeline:** gofmt, golangci-lint, GoSec SAST, SonarQube, Go race detector unit tests, MongoDB/RabbitMQ integration tests, E2E system tests, govulncheck, Docker Hub multi-image publish, self-hosted continuous deployment.
- **Complete Kubernetes Suite:** 10-manifest K8s deployment (Namespace, MongoDB StatefulSet, Services, Deployments, Ingress, ConfigMaps/Secrets, HPA, PDB).

Equifax Keying & Linking (K&L) Identity Resolution Platform

Golang, Google Cloud Spanner, GCP Pub/Sub, Apache Beam, gRPC, GCS, Jenkins CI/CD, FCRA Governance

- Engineered core identity resolution algorithms and asynchronous event-driven address linking pipelines on GCP under strict FCRA data purposing and privacy governance rules, processing millions of consumer identity records.

Clay Works IoT & Workspace SaaS (QuantiqueMinds)

Python (FastAPI), Golang, MongoDB, Redis, MQTT, AWS IoT Core, Docker, CI/CD, Agentic AI

- Engineered asynchronous FastAPI microservices and telemetry ingestion pipelines on MongoDB/Redis for real-time device control; integrated Python predictive AI workflows for workspace occupancy and energy optimization across 500+ IoT devices.

Go-STP Distributed Telecom Signaling Platform (Monty Mobile)

Golang, C++, PostgreSQL, Keycloak, SS7, STP

- High-availability signaling transfer point engine handling high-volume telecom transaction routing, global roaming, and Multi-IMSI product support with a 35% throughput improvement.

EDUCATION

Master of Computer Applications (MCA)

Utkal University, Orissa

Bachelor's Degree

Utkal University, Orissa

Diploma in Civil Engineering

SCTE&T, Orissa

CERTIFICATIONS & CONTINUOUS LEARNING

- **Yuva AI for All** — Nasscom FutureSkills Prime (2026) *(Credly Verified)*
- **Introduction to Modern AI** — CISCO Networking Academy (2026)
- **Cloud-Native & Emerging Technologies** — Participation Certificate (2026)

